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كلية تكنولوجيا المعلومات

MAMMO AI

(EARLY DETECTION OF BREAST CANCER USING A FINE-TUNED VGG16 CONVOLUTIONAL NEURAL NETWORK)

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[RAMA]

"The first people I would like to thank and express my deepest gratitude to are my dear parents, the most important people in my life. My mother, who never gets tired of praying for me and who always lights my way no matter how dark life becomes. With her encouragement, words, and advice, she makes every step easier and every dream seem closer.

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[NERMEEN]

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[YASMIN]

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As the saying goes: “The pain you feel today is the strength you will gain tomorrow.”

Abstract

Breast cancer is the most common cancer among women worldwide and a leading cause of cancer-related deaths, ranking as the second most common cause of death among females. Statistics show that 1 in 28 women is likely to develop breast cancer during their lifetime, with over two million new cases diagnosed globally each year. In Jordan, the high-risk age group is between 43 and 46 years, while in Western countries, it is more common between 53 and 57 years, although there is currently no complete cure, early detection and diagnosis greatly increase the chances of survival, making timely and accurate screening essential. However, distinguishing between benign and malignant tumors remains a major challenge.

This project aims to improve early breast cancer detection using machine learning models on breast imaging datasets. By applying advanced techniques such as feature selection and model optimization, the study addresses common challenges like low accuracy, over-fitting, and poor generalization in current models. A comparative analysis of different detection methods will be conducted to evaluate the performance of the proposed approach. The goal is to develop a more accurate, reliable, and clinically useful model that supports early diagnosis and improves treatment outcomes. The model's performance will be assessed using key diagnostic metrics to demonstrate its advantage over existing baseline models. Additionally, raising awareness about modern and diverse detection methods remains crucial to saving lives.

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List of Abbreviations

Acronym	Meaning
CNN	Convolutional Neural Network
ML	Machine Learning
DL	Deep Learning
CAD	Computer-Aided Diagnosis
SVM	Support Vector Machine
RF	Random Forest
VGG16	Very Deep Convolutional Network with 16 layers
F1	F1 measure (harmonic mean of precision and recall)

CHAPTER ONE

INTRODUCTION

1.1 Overview

Breast cancer starts in the breast tissue when abnormal cells grow and multiply out of control. It is the most common type of cancer among women in both developed and developing countries. In 2011, breast cancer caused around 508,000 deaths worldwide, and in 2012, it made up 25% of all cancer cases and 15% of cancer-related deaths among women [1]. Cancer in general puts a heavy burden on healthcare systems around the world [8]. When cancer is found at a late stage, it often leads to lower survival rates, higher treatment costs, and more deaths. On the other hand, early detection greatly increases the chances of successful treatment and long-term survival [1] [2].

Thanks to the growing availability of open-source medical datasets and advanced tools, technology now plays a big role in improving how accurately and quickly diseases are diagnosed [4] [9]. As a result, doctors are increasingly using machine learning to help make better medical decisions and improve patient outcomes [5] [3]. This project uses publicly available mammogram image datasets [10]and builds code to create a more effective system for detecting breast cancer early.

1.2 Problem Statement

Although machine learning models for breast cancer detection already exist, many of them face significant challenges such as low accuracy, overfitting, and poor generalization to new or unseen data [4] [2]. These limitations can lead to delayed

or incorrect diagnoses, which may negatively impact patient outcomes [3]. In addition, some models struggle to distinguish between subtle differences in benign and malignant tumors, reducing their diagnostic reliability [6]. The current approaches often lack the precision, robustness, and consistency required for real-world medical use [7]. This highlights the urgent need for improved models that can offer higher diagnostic accuracy, better generalization across diverse patient populations, and reduced error rates in early detection [8] [9]. Enhancing these aspects is critical for developing tools that can truly assist healthcare professionals and support timely, life-saving interventions.

1.2 Project objectives

The main objectives of this project are:

- To initiate the study by conducting an extensive review of related research and prior studies in order to identify existing challenges and gaps in breast tumor classification using mammogram images [2] [3].

- To develop and investigate an advanced deep learning framework for the accurate classification of breast tumors (malignant or benign) based on mammogram images [4].

- To significantly enhance the detection rate and generalization capability of Convolutional Neural Networks (CNNs) by leveraging the VGG16 architecture through transfer learning, combined with robust data augmentation techniques.

- To rigorously evaluate the performance of the proposed method using comprehensive and well-established diagnostic metrics (accuracy, precision, recall, and F1-score), thereby demonstrating its clinical utility and superiority over existing baseline models [4] [7].

- To validate the effectiveness of the optimized VGG16-based framework through comparative analysis with conventional CNN approaches [6] [11].

1.4 Research framework

This research follows a systematic approach to improve the performance of a breast cancer detection model using mammogram images. The goal is to enhance the classification accuracy of the existing model through algorithmic improvements and careful model development [4] [2].

The research strategy includes the following steps:

- Data Collection and Preprocessing: The mammogram dataset will be prepared by addressing issues.
- Model Selection and Optimization: The current model reviewed and optimized by focusing on extracting the most relevant features from the images and improving its ability to detect patterns linked to breast cancer [12] [13] [14].
- Evaluation and Testing: The model assessed using standard performance metrics, such as accuracy, precision, recall, and F1-score, to evaluate its effectiveness [4] [7].
- Iterative Refinement: Based on the evaluation results, the model undergoes iterative refinements. This may include adjusting the model architecture,

retraining with different feature sets, or applying advanced optimization techniques to improve overall performance and reliability [5] [6].

This approach aimed to produce a more accurate and dependable system for early breast cancer detection using mammogram images [8] [3].

1.5 Scope (Boundary)

This section delineates the boundaries and limitations of the project, defining what is included within its objectives and what falls outside its immediate focus. It outlines the specific aspects of breast cancer detection that this research addresses, the types of data utilized, and the expected outcomes, ensuring a clear understanding of the project's reach and constraints.

1.5.1 Scope Description

This project aims to enhance the accuracy and reliability of breast cancer detection through the analysis of mammogram images. It focuses on improving an existing machine learning model that is used to classify breast tumors into two categories: benign (non-cancerous) and malignant (cancerous). In order to achieve more accurate and clinically meaningful results, the project incorporates several advanced techniques such as feature selection, data preprocessing, and model optimization. These enhancements are expected to significantly improve the model’s performance by reducing false positives and false negatives, which are critical in medical diagnosis. Ultimately, the goal is to develop a more robust and trustworthy system that can support healthcare professionals in making more informed and accurate diagnostic decisions.

1.5.2 Deliverables

Improved Machine Learning Model: A robust and accurate deep learning model capable of classifying breast tumors (benign or malignant) from mammogram images, specifically leveraging the fine-tuned VGG16 architecture. Source Code: The complete Python source code for the project, encompassing scripts for data preprocessing, model training, evaluation, and the implementation of the fine-tuned VGG16 CNN architecture. Interactive Desktop Application: A user-friendly desktop application designed to display the classification results, providing an interactive environment for medical professionals. Performance Comparison Report: A detailed report presenting a comparative analysis of the proposed model against a baseline (previous) algorithm, utilizing the same mammogram dataset to quantitatively demonstrate the significant improvement achieved by the enhanced VGG16-based approach.

1.5.3 Acceptance Criteria

The success of the project determined by meeting the following specific, measurable, and testable acceptance criteria [4] [3] [7]:

- Model Performance:
  - The fine-tuned VGG16 model must achieve a classification accuracy of at least 98% on the validation dataset.
  - The model must demonstrate a precision and recall of at least 98% for the 'Malignant' class to minimize false negatives.
- Application Functionality:
  - The desktop application must successfully load a mammogram image selected by the medical professional.

    - The system must display the classification result (Benign or Malignant) along with a confidence score within .34 seconds of image submission.
- Usability:
    - A new user should be able to perform a classification task (upload image and get a result) without prior training or assistance, based solely on the on-screen instructions.

1.5.4 Methodology

The project followed the agile methodology (As shown in Chapter 3) for time management and executing different stages, allowing for continuous development and evaluation of the model based on results.

1.6 Gantt chart

Figure 1.1 shows a Gantt chart that visually represents the timeline of various project tasks from March to August. Each row corresponds to a specific task, and the colored bars indicate the duration and timing of each activity. The tasks include project planning, research, system implementation, testing, evaluation, presentation, and submission. This chart helps organize and track the progress of each task. The first three phase's project planning, research, and system implementation were completed during the initial three months, from March to May. The remaining phases, including testing, evaluation, presentation, and submission, are scheduled to be completed over the following three months.

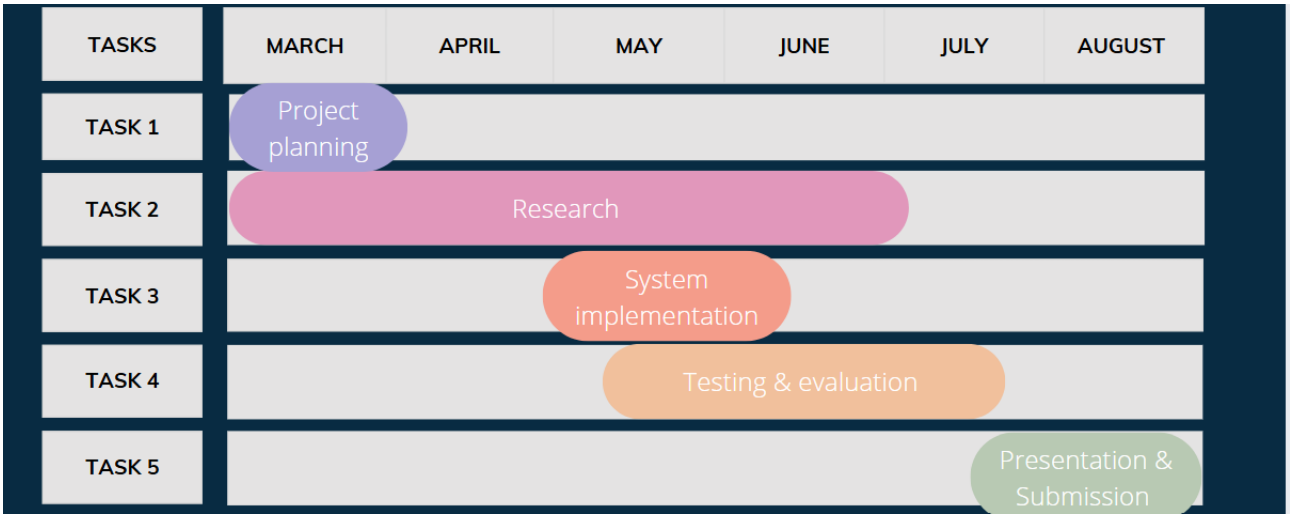

Figure 1.1: Gant Chart

1.6 Project outline

This report is structured into six main chapters, each detailing a specific aspect of the breast cancer detection project:

- Chapter 1: Introduction

Provides a comprehensive overview of breast cancer, highlighting its global and local prevalence (specifically in Jordan), its critical importance as a public health issue, and the urgent need for early detection. It also discusses the inherent challenges in accurately distinguishing between benign and malignant tumors, setting the context for the project.

- Chapter 2: Literature Review

Presents a thorough review of existing research and methodologies in breast cancer detection using machine learning and deep learning techniques. This chapter critically evaluates common approaches, identifies current limitations, and highlights research gaps, thereby justifying the proposed methodology.

- Chapter 3: Methodology

Details the systematic approach undertaken in this project. It describes the dataset utilized, the comprehensive data preprocessing and augmentation strategies employed, and the specific machine learning models and advanced techniques applied, with a particular focus on the fine-tuned VGG16 architecture and transfer learning, to enhance detection accuracy.

- Chapter 4: Design Models

Outlines the architectural design of the system. This chapter includes detailed diagrams and descriptions of various components, illustrating how different parts of the system interact and how data flows through them to achieve the project objectives.

- Chapter 5: Experiments and Results

Presents the experimental setup, procedures, and the quantitative results obtained from testing the developed model. This chapter includes a detailed analysis of the model's performance, a comparison with other relevant models, and a clear demonstration of the improvements achieved.

- Chapter 6: Conclusion and Future Work

Summarizes the key findings and achievements of the project, reiterating its significance and contributions. It also provides recommendations for future research directions and potential enhancements to the current system.

CHAPTER TWO

LITERATURE REVIEW

2.1 Overview

The purpose of this literature review is to explore existing research on using machine learning and deep learning for breast cancer detection. It looks at important parts of past studies, such as how the data is prepared, how models are chosen, how features are extracted from images, and how the models are evaluated [2] [3]. This review also highlights the strengths and weaknesses of previous work, helping to shape the direction of this project [4].

Technology in the medical field is growing rapidly, leading to the creation of advanced tools that improve disease diagnosis, treatment planning, and many other areas like drug development, medical imaging, and electronic health records [8]. One important development is the use of Computer-Aided Diagnosis (CAD) systems. These systems help doctors, especially radiologists, by giving a second opinion when reading mammograms. CAD systems are being developed to tell the difference between harmless and harmful breast tumors. They often use machine learning and deep learning methods to detect small patterns in medical images that can be hard for the human eye to see [4] [7].

2.2 Critical evaluation (gap analysis)

Many machines learning models have been developed for detecting breast cancer, but they often have problems when used in real-world medical settings [3]

[7]. Some are trained on small or unbalanced datasets, which can cause them to make mistakes when analyzing new data [2]. Also, many of these models are hard for doctors to understand or use This project aims to solve these issues by improving an existing CNN model using feature selection and optimization [4] [8].

2.3 Related Work

A lot of research has looked into using artificial intelligence to detect breast cancer, especially by analyzing medical images [3] [4]. Some studies use traditional machine learning methods like Support Vector Machines (SVM) and Random Forests (RF) [2], while others use more advanced deep learning models like Convolutional Neural Networks (CNNs) [7] [12]. Each method has its own strengths, such as good accuracy, the ability to handle large amounts of data, and support for automatic detection [9].

Table 2.1: Summary of Related

Study title	Year	Methodology	Key finding	Limitation
Update on US Breast Cancer Statistics [15].	2022	**Population-based analysis of incidence, mortality, survival, and screening data.**	The mortality rate decreased by 43% between 1989 and 2020.	Data limitations include variations in data sources, estimation of figures, and some data being incomplete or based on reported information.
Ensemble Learning Techniques for	2021	**Stacking, Bagging, Boosting**	Ensemble methods improved	Model complexity increases

Cancer Detection [17]			classification performance	
Deep Learning Applied for Histological Diagnosis of Breast Cancer [18]	2020	**ResNet50 and DenseNet121**	The models achieved high performance	The BreakHis dataset is limited to 82 patients and shows lower performance at 400x magnification compared to other levels.
Deep Breast Cancer Net: Ultrasound Model [19]	2023	**Deep Breast Cancer Net on US images**	Accuracy ~98%	focused on ultrasound data only; limited generalizability

Table 2.1 summarizes various studies related to breast cancer detection and classification methods. It includes details on the utilized methodologies, key findings, and limitations of each study. This table provides an overview of the advancements and challenges in applying machine learning and deep learning techniques for breast cancer diagnosis.

2.3.1 Discussions of the problem

Several studies have explored breast tumor classification using deep learning models such as Convolutional Neural Networks (CNNs). These models have achieved promising accuracy levels, often above 80% [18], but their performance can decline when trained on imbalanced datasets. Researchers have also applied transfer learning, using pretrained models like VGG16, to improve accuracy, especially when data is limited [12]. Data preprocessing and augmentation play important roles

in enhancing model generalization and preventing overfitting. However, many existing studies focus mainly on model development and do not provide a complete system that includes user-friendly tools for medical professionals.

2.3.2 Issues and challenges faced by the other project

One common issue in previous studies is overfitting, which often happens due to small datasets or weak validation methods. This results in models that perform well during training but struggle with real-world data. Additionally, many models act as black boxes with little transparency, making doctors hesitant to trust them in clinical settings. The lack of standard preprocessing steps like uniform normalization and data augmentation makes it difficult to compare results across studies. Additionally, many projects overlook the user experience, failing to consider how medical professionals will interact with the system. These challenges shaped the direction of this project. To address them, techniques such as robust data augmentation and cross-validation were applied to improve the model's reliability and generalizability. Moreover, the system is deployed as a standalone executable file with clear and intuitive user interfaces, ensuring accessibility and ease of use for medical professionals.

2.4 Summary

This chapter provides an overview of previous studies that used machine learning and deep learning techniques for breast cancer detection through mammogram images. These studies demonstrated that models such as Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs) can help improve diagnostic accuracy.

However, the literature reveals several research gaps. First, many models were trained on small or imbalanced datasets, which limits their generalizability in real-world medical settings. Second, most studies focused either on medical images or clinical data, without integrating both types of information to achieve a more comprehensive diagnostic approach.

Additionally, many of the previous models lacked user-friendly interfaces, making it difficult for non-technical medical professionals to adopt them in clinical practice. Furthermore, some research failed to evaluate the practical applicability of the models outside academic environments, particularly regarding prediction speed and classification accuracy in borderline cases. These gaps highlight the need for further studies that use more balanced datasets, and consider practical implementation aspects to ensure usability in real-world healthcare systems. Based on the foregoing, it is evident that there is a persistent need to improve the accuracy of breast cancer detection models and reduce their susceptibility to overfitting. To address these challenges observed in previous studies, our project adopts an advanced methodology based on Transfer Learning using the VGG16 architecture. Furthermore, extensive Data Augmentation techniques will be applied, and regularization layers such as Dropout will be utilized. These are strategies aimed directly at enhancing the model's ability to generalize and achieve more stable and accurate results.

CHAPTER THREE

METHODOLOGY

3.1 Overview

This chapter presents the methodology used to develop the breast cancer detection project. It explains the overall approach and the steps followed to achieve the objectives of the project. The methodology outlines the stages of data collection, preprocessing, model development, evaluation, and deployment.

The used approach ensures a systematic and organized workflow, allowing for continuous improvement and validation of the model's performance. By following a clear methodology, the project aims to deliver a reliable, accurate, and user-friendly system for the breast cancer detection that meets the defined functional and non-functional requirements.

3.2 Feasibility Study

A comprehensive feasibility study was conducted to assess the viability of this project from technical, operational, and economic perspectives. This analysis ensures that the proposed solution is not only innovative but also practical and sustainable.

3.2.1 Technical Feasibility:

The project relies on established and widely supported deep learning frameworks (TensorFlow, Keras) and programming languages (Python) [20]. The computational resources required for training advanced CNN models like VGG16 are readily available through cloud computing platforms or high-performance local machines. Furthermore [12], the availability of well-documented public datasets [10] for breast cancer imaging confirms the technical possibility of data acquisition and processing. The integration of transfer learning [5] significantly reduces the need for extensive computational power and vast proprietary datasets, making the technical implementation highly feasible.

3.2.2 Operational Feasibility:

The proposed system is designed to seamlessly integrate into existing clinical workflows, providing an intuitive and user-friendly interface for medical professionals. By automating the preliminary classification of mammogram images, the system aims to reduce the workload on radiologists, minimize human error, and expedite the diagnostic process. The output (benign/malignant classification with confidence scores) is directly actionable and complements current diagnostic practices, enhancing decision-making without requiring significant changes to operational procedures. The system's ease of use, as highlighted in the acceptance criteria, ensures its practical adoption [8].

3.2.3 Economic Feasibility:

The economic benefits of early breast cancer detection are substantial, far outweighing the development and operational costs of this system. Early diagnosis leads to less invasive treatments, reduced healthcare expenditures, and significantly improved patient outcomes and survival rates [1]. While initial investment in computational resources and development time is required, the long-term societal and economic returns, including saved lives and reduced treatment burdens, justify the project. The use of open-source libraries and pre-trained models (VGG16) also helps in minimizing development costs.

3.3 Risk Analysis

Table 3.2 express the potential problems (risks) might occur during the process of breast cancer detection. For each problem, we explain what it is, how likely it is to happen, how serious it could be, and what we can do to avoid or fix it. The table contain these parts:

a. Risk Type: The kind of problem (for example: technical error, data loss, or delays).
b. Description: A short explanation of the problem and how it could affect the system.
c. Probability: How likely the problem is to happen (Low, Medium, or High).
d. Impact: How much damage or effect the problem could have (Low, Medium, or High).
e. Mitigation Plan: What we will do to prevent the problem or reduce its effect.

Table 3.2: Risks [21]

Risk type	Description	probability	Impact	mitigation plan
Technical Risk	Issues with the Al model's accuracy or failure to correctly classify images	Medium	High	Improve data quality, regularly update the model, and conduct thorough testing before deployment
Data Loss	Risk of losing images or diagnostic results during system use	Low	High	Implement automatic local data saving .and backup functionality
Large mammogram images	Risk of system slowing down or crashing due to large file sizes	Medium	Medium	use trusted tools and test the system step by step
small or unbalanced dataset	Risk of model underfitting or bias due to insufficient data	High	High	using data augmentation and good validation methods.
time delays	Delays in project timeline due to complexity or lack of resources	Medium	Medium	follow a clear plan and divide the work into small tasks

3.4 Methodology process

In this project, the Agile methodology was adopted to manage time and guide the execution of different development stages. Agile offers flexibility, continuous feedback, and iterative development, making it ideal for projects like the proposed one, where regular updates, user involvement, and rapid adjustments are essential. The Scrum framework was specifically applied, allowing the division of work into manageable sprints, conducting regular meetings to evaluate progress, and adapting quickly to any challenges or changes in requirements. This methodology was chosen because it suits the nature of the project, where the system's purpose is clear from the beginning, but improvements and refinements are expected throughout the development process. The goal was to build a stable, functional, and user-oriented breast cancer detection system based on Artificial Intelligence.

The development process was divided into the following key phases:

- Planning:
  The project idea was discussed, and clear goals were set. The team identified the problem of manual breast cancer diagnosis and decided to build a system that leverages AI to assist in early detection.
- Requirements Analyses:
  Requirements were gathered through interviews and questionnaires with doctors and potential users. Both functional and non-functional requirements were documented to ensure the system meets clinical needs and expectations (discussed more in 3.5).

- System Design:

  We will design the system architecture, dataset, and the AI model framework. Additionally, we will plan the design of the user interface and the flow of data through the system.

- Implementation:

  The proposed model was implemented using the Python programming language and the deep learning libraries Keras and TensorFlow. The implementation phase focuses on building and training a Convolutional Neural Network (CNN) model based on the VGG16 architecture [5] [12] [20], applying the Transfer Learning technique to achieve high performance. Data Preprocessing and Augmentation: Before feeding the images into the model, they were resized to a uniform dimension of (224x224) pixels [4] [22], which is the standard input size expected by the VGG16 architecture. To address the challenge of a limited dataset size and to prevent overfitting, extensive Data Augmentation techniques were applied to the training set. These techniques included random rotations, width and height shifts, shearing, zooming, and horizontal/vertical flips. These operations artificially increased the diversity of the training data, thereby enhancing the model's ability to learn robust features and generalize to new, unseen data. Model Architecture: The model was built upon the VGG16 architecture, pre-trained on the ImageNet dataset. A Fine-Tuning strategy was employed as follows: Base Model: The VGG16 model was loaded without its top fully connected layers. The weights of most of its convolutional layers were frozen to leverage the pre-learned features (such as edges, shapes, and textures).

Custom Top Layers: On top of the VGG16 base, new layers specific to our classification task were added. This starts with a Flatten layer to convert the 2D feature maps into a single vector, followed by two Dense layers (with 128 and 64 neurons, respectively) using the ReLU activation function. In between these layers, Dropout layers (with rates of 0.5 and 0.3) were inserted. Dropout is a crucial regularization technique that reduces overfitting by randomly ignoring a fraction of neurons during training. Output Layer: The model culminates in a final Dense layer with two nodes and a SoftMax activation function, which classifies the images into two categories: Benign or Malignant. Model Training: The model was trained using the Adam optimizer with a low learning rate of 1e-5 [4] [13] [14]. This is an effective choice for fine-tuning as it ensures that the pre-trained weights are adjusted gradually. To prevent training for longer than necessary, an Early Stopping mechanism was used. This technique monitors the model's performance on the validation set and stops the training process when the performance no longer improves, restoring the best-performing weights.

- Testing:

  The system underwent multiple testing stages, including unit testing, system testing, and user acceptance testing, to ensure functionality and accuracy in a clinical context [23].
- Deployment:

  The system should be deployed on desktop. As it runs offline, it doesn’t require an internet connection, ensuring security.
- Maintenance:

  The project has not been implemented in a medical environment yet. It

should be implemented first to identify weaknesses and errors so that it can be updated and maintained.

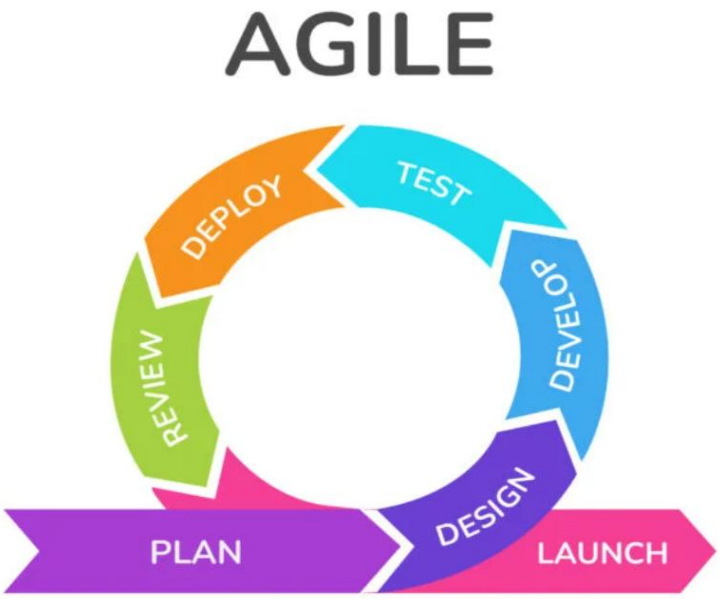

Figure 2.4.2: Agile Software Lifecycle

3.5 Requirements

Artificial Intelligence (AI) can help detect breast cancer at an early stage. Several articles were reviewed to learn about new methods in medical imaging and machine learning [4] [8]. To ensure the project is useful in real life, doctors were consulted about their needs. Their answers provided insights into the real problems faced in breast cancer diagnosis and guided the development of the system.

3.5.1 Requirements Elicitation

The requirements elicitation process was carried out using three main methods: literature review, interviews, and surveys. The literature review

helped in identifying system features, performance benchmarks, and best practices from previous studies. Interviews with experienced doctors provided valuable insights into real challenges in breast cancer diagnosis and guided the refinement of requirements to fit clinical needs. In addition, an online survey collected feedback from users about system design, usability, and accuracy expectations. The combination of these methods ensured that the requirements are both scientifically valid and practically relevant to real-world applications.

3.5.1.1 Literature Review:

The literature review was conducted to explore previous research studies related to breast cancer detection using artificial intelligence, especially machine learning and deep learning techniques. By analyzing existing studies, we identified common system features, performance benchmarks, and best practices that helped in shaping the initial system requirements [5] [9] [12] [24]

3.5.1.2 Interviews with Doctors:

Interviews were conducted with a group of doctors specialized in breast cancer diagnosis. The interview questions were prepared under the supervision of faculty members to ensure their academic and clinical relevance. The aim of these interviews was to identify the practical challenges faced by doctors in clinical settings and to gain in-depth insights into their needs and expectations from an AI-based detection system. These interviews significantly contributed to refining the system requirements to ensure compatibility with real-world clinical applications and ease of use.

It is worth noting that the interviewed doctors were specialists with extensive experience in breast cancer diagnosis, which added a high level of scientific credibility and relevance to the collected data.

For the full list of interview questions, refer to Appendix A.

3.5.1.3 Survey:

An online survey [13] was designed to collect user feedback on the system's design, usability, and the essential requirements expected from a breast cancer detection system. The survey included ten questions covering expected features, required accuracy levels, user interface preferences, and the importance of early detection. A total of 41 participants responded to the survey, and their feedback was carefully analyzed. These insights played a significant role in guiding the system's redesign and improvement to better meet the practical needs of clinical environments. These results indicate a high level of awareness and a significant degree of acceptance toward the use of AI applications in early breast cancer detection, although some concerns remain regarding the accuracy of such systems.

It should be noted that the survey questions were developed under the supervision of a committee of specialized faculty members to ensure the validity, accuracy, and relevance of the questions to the research domain.

For the full list of survey questions, refer to Appendix B.

Based on the survey results, the following key statistics were observed:

• 97.6% of participants were aware of Artificial Intelligence (AI).

• 68.3% believed that AI could assist in the early detection of breast cancer.

• 70.7% trusted AI-based diagnostic systems.

• 75.6% felt comfortable using AI as a diagnostic aid [1].

3.5.2 Functional Requirements:

3.5.2.1 The system must allow the uploading of a mammogram.
3.5.2.2 The system must process the image using a trained artificial
3.5.2.3 intelligence model (such as a CNN).
3.5.2.4 The system must classify the image as benign or malignant.
3.5.2.5 The final result must be clearly displayed to the user (with a confidence level).

3.5.3 Non-Functional Requirements:

3.5.3.1 Performance: The analysis process should take no more than a few seconds.
3.5.3.2 Security: Patient data should be encrypted, and results should only be shared with authorized users.
3.5.3.3 Usability: An easy-to-use interface for users with non-technical medical backgrounds.
3.5.3.4 Accuracy: The model's prediction accuracy must exceed 98%.
3.5.3.5 Stability: The system must operate without crashes.

3.6 System design

- Architectural Model: The system is designed as a standalone application, packaged into an executable file that runs locally on the user's device. All processing, including AI-based image analysis, is performed on the client side without requiring a server connection [23].
- Database: A local database or file-based storage is used to store image analysis results and reports [8].
- System Modules: Image Uploading Module, Image Analysis Module (CNN model) [12], Results Display Module, Report Generation Module [22] [25].

3.7 Implementation

The model was implemented in Python using deep learning libraries such as Keras and TensorFlow [20]. Training was conducted on the Mammogram Mastery: A Robust Dataset for Breast Cancer Detection and Medical Education dataset to classify breast tumors [10]. The user interface was developed using Tkinter to provide a simple and accessible desktop application [23]. An SQLite database was integrated to store patient information, diagnosis results, and image analysis records. The system was converted into a standalone desktop application using tools like PyInstaller or auto-py-to-exe, allowing it to run independently on local machines without requiring a Python environment.

3.8 Testing:

- Model Testing: The data was split into training and test sets to verify accuracy and effectiveness.

- Unit Testing: Each module (image upload, result display) was independently verified.
- Integration Testing: Ensure all modules work seamlessly together.
- User Acceptance Testing: The system was introduced to a number of users (doctors) for testing.

- Success Criteria: Model prediction accuracy exceeding 90 or more, Response speed within seconds, Ease of use and error-free operation [4].

3.9 Deployment:

The system runs locally (offline) and directly on the doctor's device. The following steps were implemented:

- Device Preparation: project was installed the necessary software and tools to run the system (such as Python and the required libraries).
- System Installation: project was provided an easy-to-use setup file that can be run on any device.
- System Testing: After installation, the system was tested on devices to ensure it worked properly.
- User Guide: A simple file explaining how to use the system step-by-step was prepared to help doctors operate it without problems.

This deployment type was chosen to maintain privacy and ease of use, as no data is sent over the internet [8].

3.10 Maintenance

The project has not been implemented in a medical environment yet. It should be implemented first to identify weaknesses and errors so that it can be updated and maintained.

CHAPTER 4

DESIGN MODELS

4.1 Overview

This chapter delves into the architectural and functional design of the proposed breast cancer detection system. It presents a series of design models and diagrams that visually represent the system's structure, behavior, and data organization. Each model serves a specific purpose in elucidating how the system interacts with its environment, processes data, and fulfills its objectives. These diagrams collectively provide a comprehensive blueprint for the system's implementation, ensuring clarity, consistency, and a shared understanding among stakeholders. The models presented include context diagrams, data flow diagrams, use case diagrams, and entity-relationship diagrams, all tailored to illustrate the robust design supporting the VGG16-based detection framework.

4.2 Context diagram-0.

Figure 4.2.3 shows a context diagram of the Breast Cancer Detection System. and its interactions with external entities. It depicts the system as a single process, highlighting the primary inputs it receives and the outputs it produces, along with the external entities that interact with it. This diagram is crucial for defining the system's boundaries and understanding its scope within the broader environment.

In the system, the primary external entities include:

Medical Professionals: These users provide mammogram images as input to the system and receive classification results and confidence scores as output. They are the primary beneficiaries of the system's diagnostic support.

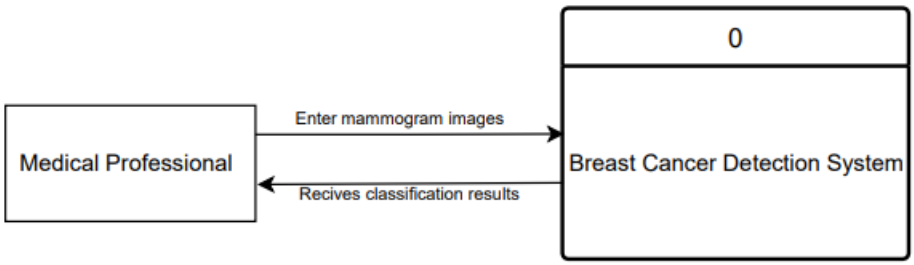

Figure 4.2.3: Context Diagram-0

4.3 Data flow Diagram (DFD)-1

Figure 4 shows the (DFD) of the Breast Cancer Detection System. It details the main processes, data flows, and stores inside the system.

- Medical Professional uploads a mammogram image and receives the final diagnostic report.
- Process 1.3 (Preprocessing): receives uploaded images from the Medical Professional and raw images from the Image Dataset (D1). It applies preprocessing steps (resizing, normalization, enhancement) and outputs a preprocessed image.
- Process 1.1 (Augment Images): generates additional training samples from preprocessed images using augmentation techniques to improve model generalization.
- Process 1.2 (Training): uses both original and augmented images to train the CNN model and stores the trained parameters in Trained_Model_DS (D2).

- Process 1.4 (Feature Extraction and Classification): takes the preprocessed image and retrieves the trained model from D2 to extract features and classify the tumor as benign or malignant.
- Process 1.5 (Result Generation and Display): formats the classification result and delivers the diagnostic report back to the Medical Professional.

The diagram demonstrates the complete workflow, showing how mammogram images flow from upload and preprocessing, through augmentation, training, feature extraction, and classification, and finally to the generation of a diagnostic report for clinical use.

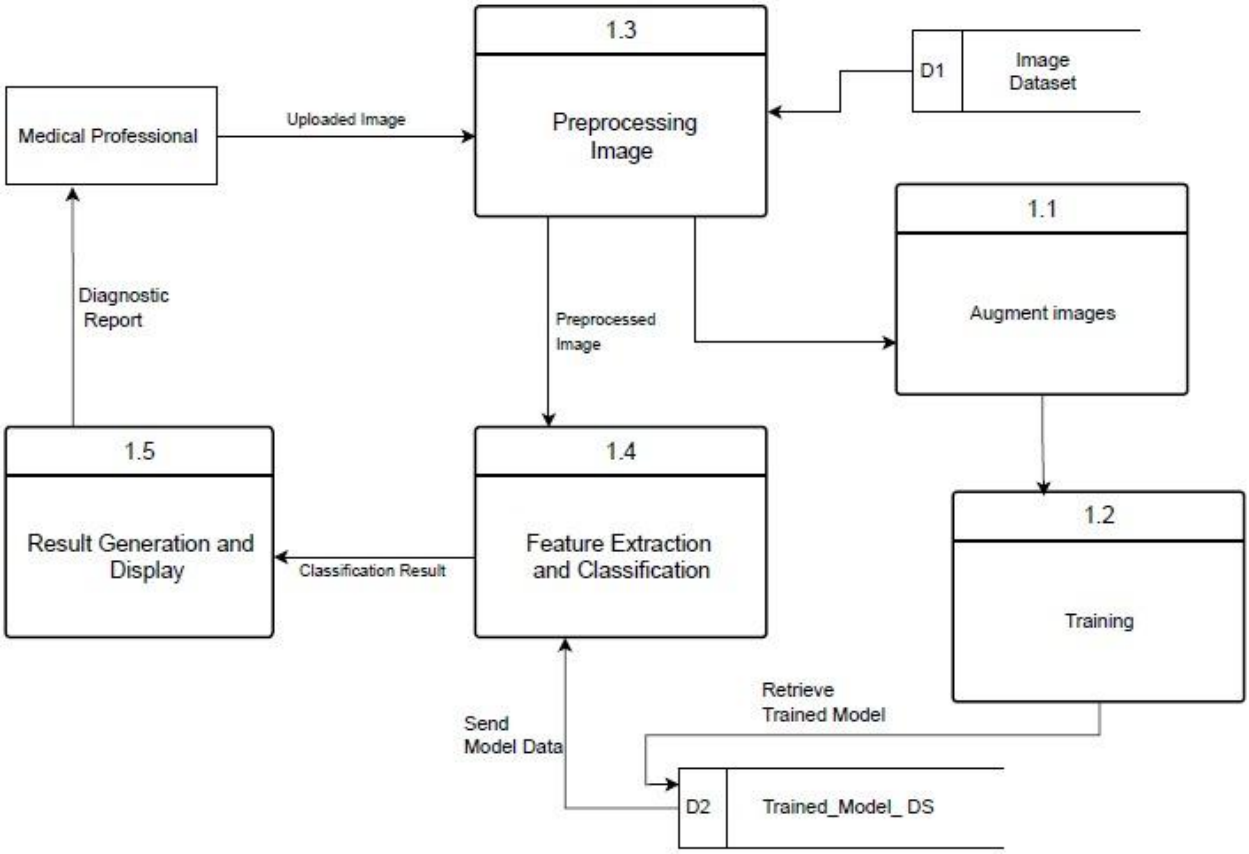

Figure 4.3.4: Level-1 Data Flow Diagram (DFD)

4.4 Use Case Diagram

Figure 4.4.5 shows a Use Case Diagram of the Breast Cancer Detection System. It illustrates the system's functionality from the perspective of its users.

- Actors:
  - Medical Professional: The primary user who initiates processes and receives the results.
  - Model: The entity that performs the internal, automated operations.
- Use Cases:
  - Upload Mammogram Image: Initiated by the Medical Professional to input the image into the system.
  - Preprocess Image: Executed by the System to prepare the image (resizing, normalization) for analysis.
  - Feature Extraction: Performed by the System to identify and extract relevant features from the image (textures, shapes, edges).
  - Classify Image: Executed by the System to categorize the image as Benign or Malignant.
  - Generate Diagnostic Report: Performed by the System to produce a report with the analysis results and present it to the Medical Professional.
- Relationships:
  - Arrows from the Medical Professional to the use cases show that the professional initiates the Upload and Generate Report actions.
  - Arrows from the system to the use cases indicate that the model is responsible for executing the Preprocessing, Feature Extraction, and Classification tasks.

- They include relationship between Classify Image and Feature Extraction is a key element. The dashed arrow from Classify Image to Feature Extraction signifies that the Classify Image use case must include or depend on the Feature Extraction use case. This means that classification cannot occur without the features first being extracted.

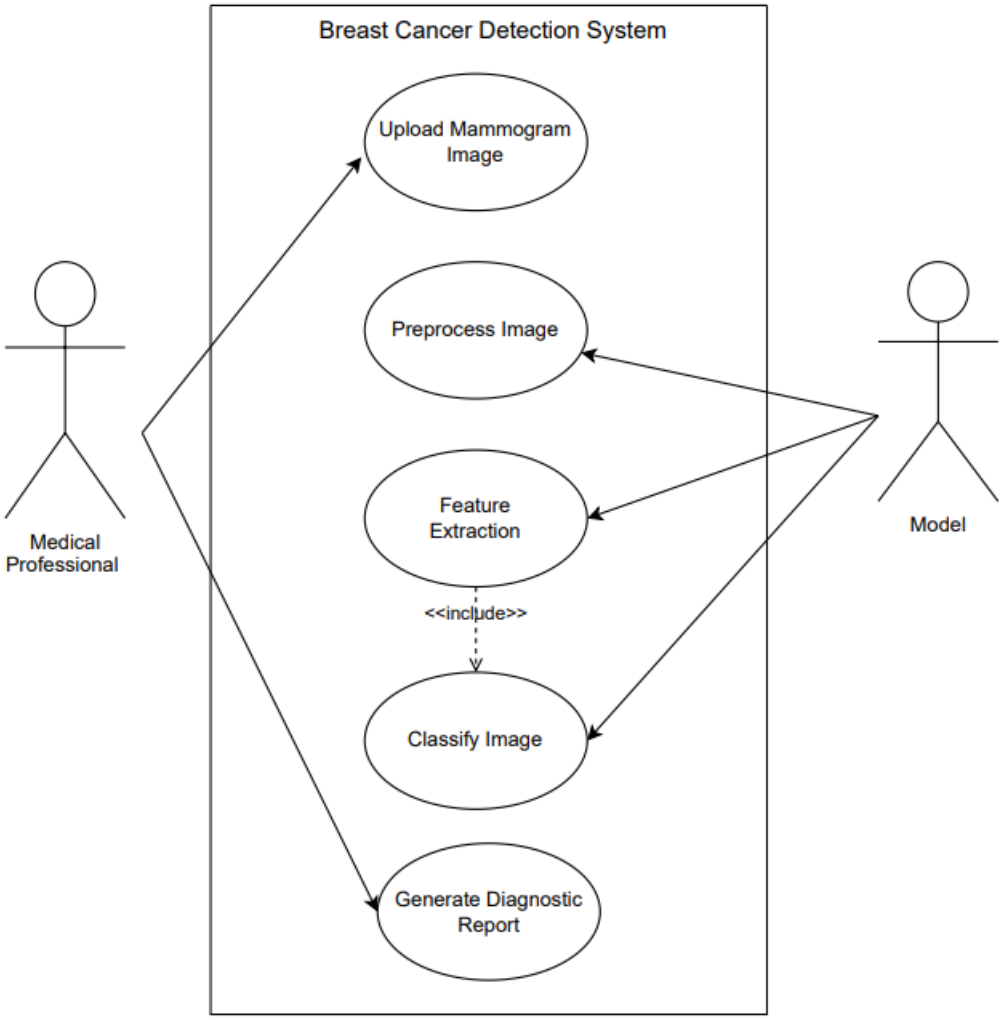

Figure 4.4.5: Use Case Diagram

4.5 Sequence diagram

Figure 4.5.6 shows the Sequence Diagram of the that illustrates the interaction flow between a user and the components of a breast cancer detection system. It shows how information is passed sequentially from the initial image input to the final result display.

- The Actor: Medical Professional initiates the process by providing the mammogram image.
- The New Detection Interface: This component receives the raw image from the medical professional. It acts as the primary point of interaction and ensures the flow starts correctly.
- Preprocessing and Segmentation: It takes the raw image and prepares it for analysis. This process involves steps like resizing, normalization, and enhancing the image to improve clarity.
- Feature Extraction: This component receives the preprocessed image and identifies crucial features (like shape, size, and texture) from potential tumors, which are vital for the diagnostic process.
- Classification: Using the extracted features, this component classifies the image, determining whether the signs point to a benign or malignant case.
- The Detection Result Interface: This is the final component in the chain. It receives the classification decision and presents a diagnostic report or a clear result to the medical professional.

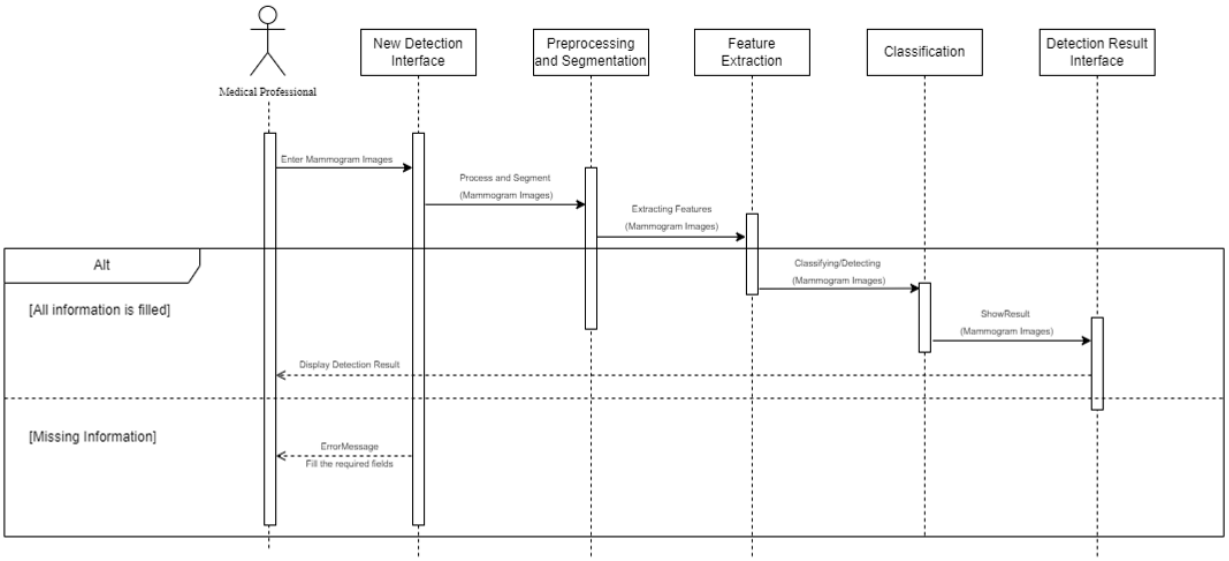

Figure 4.5.6: Sequence Diagram

4.6 Design Pattern

Design patterns are proven solutions to common software design problems. They provide a structured way to organize code, improving reusability, maintainability, and scalability of applications.

As shown in **Figure 4.6.7**, in the context of the MammoAI system, the Singleton Pattern (Creational) was applied. This pattern ensures that only one instance of a class exists throughout the program and provides a global point of access to it. In MammoAI, the Singleton Pattern is used to manage the machine learning model efficiently. The model is trained once, and its weights are saved and reused whenever needed. This prevents unnecessary retraining, saves computational resources, and guarantees consistent predictions across the application.

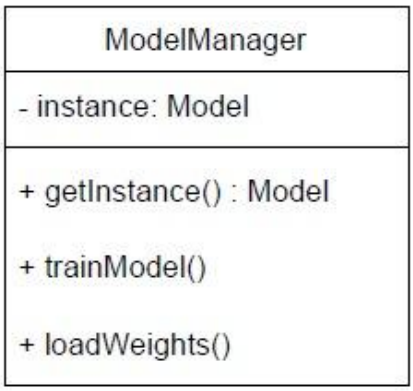

Figure 4.6.7:Singleton Pattern

4.7 Program Interface

The program interface was developed using Tkinter to provide a simple and user-friendly environment for interaction with the breast cancer detection system. It allows users to:

- Upload Medical Images: Users can browse and select mammogram images from their local device.
- Image Display: original of the uploaded image are shown.
- Prediction Result: The system displays whether the tumor is classified as benign or malignant, along with the prediction confidence.

The figures down below represent the program interfaces in order and how it work:

Figure 4.7.8 shows the license agreement window of the MammoAI system. It presents the software terms and conditions, including disclaimers about warranties, responsibilities, and medical usage restrictions. At the bottom, the user can either accept by clicking "I Agree" to continue or decline by clicking "I Disagree" to exit.

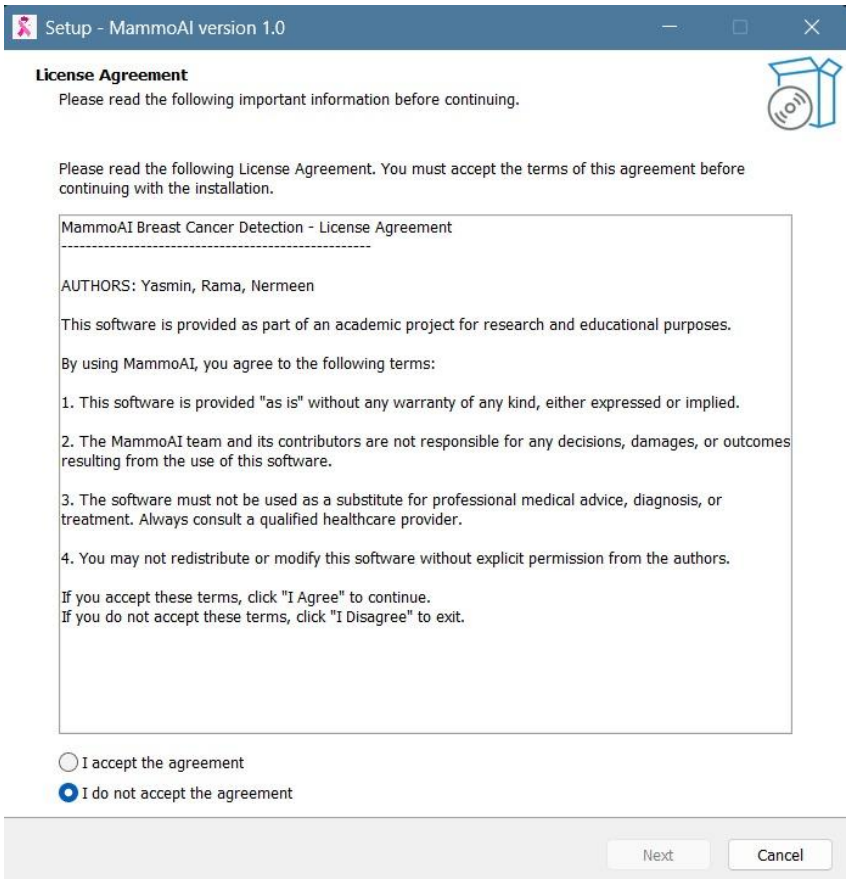

Figure 4.7.8: License Agreement

Figure 4.7.9 displays the “Select Additional Tasks” window during the installation process of the MammoAI system. In this step, the user can choose extra setup options, such as creating a desktop shortcut, before proceeding with the installation.

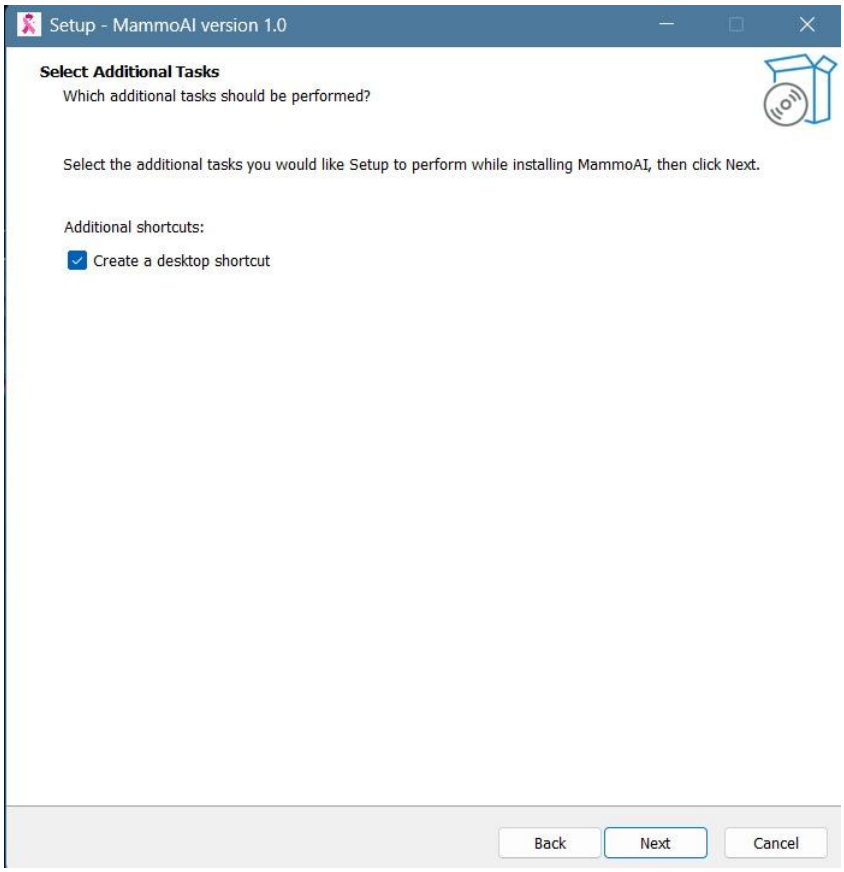

Figure 4.7.9: Select Additional Tasks

Figure 4.7.10 displays the “Ready to Install” screen, where the user can start the installation. Once the Install button is clicked, the program will begin downloading and installing.

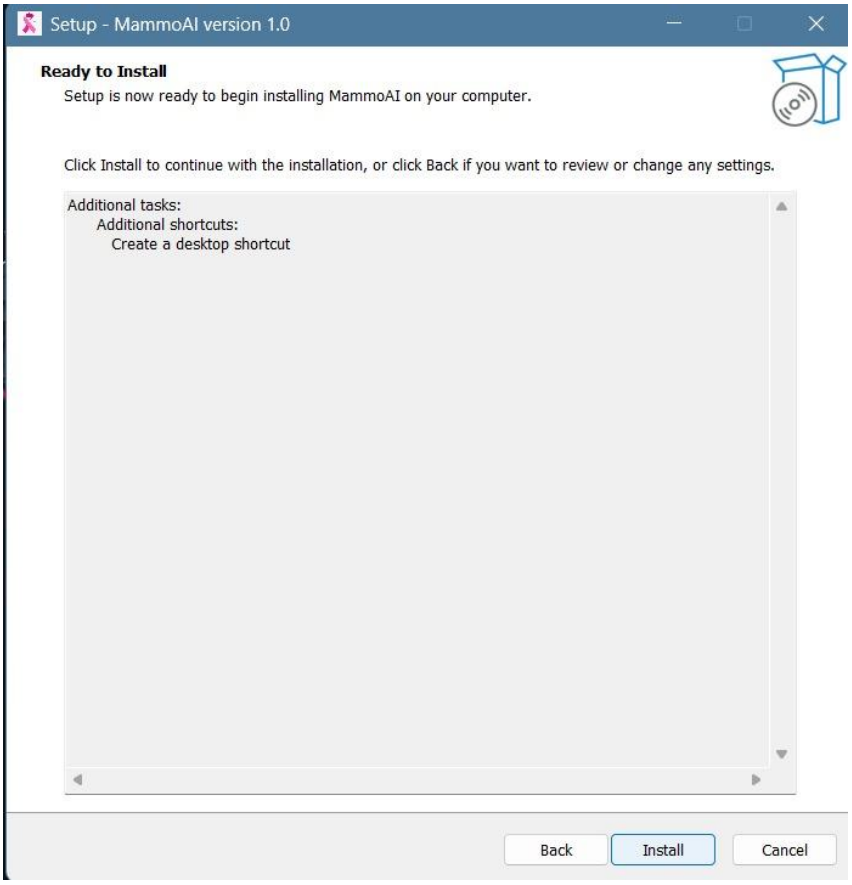

Figure 4.7.10: Ready to Install

Figure 4.7.11 shows the installation progress screen, where the user waits while the program is being installed.

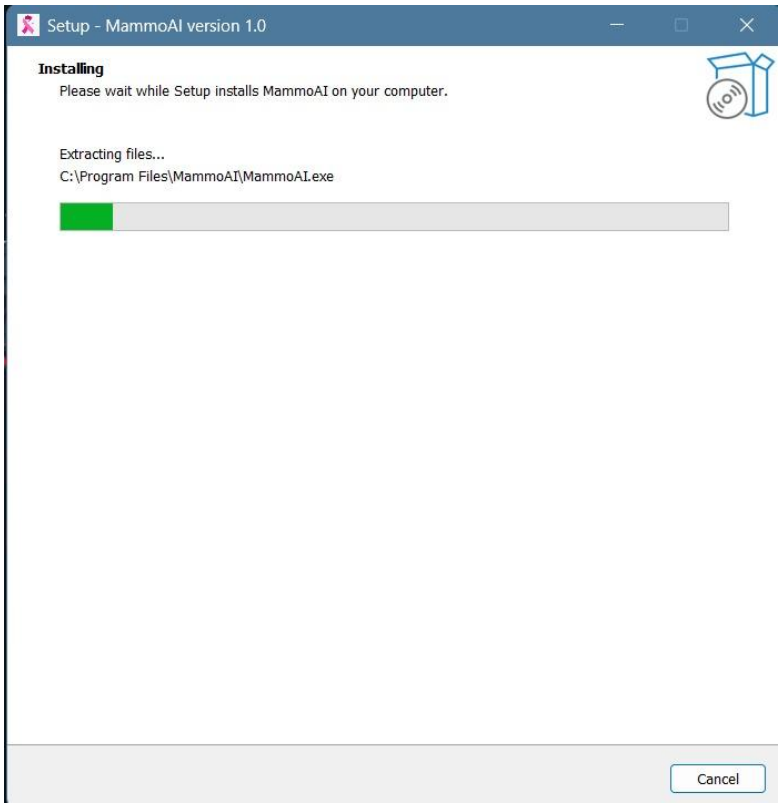

Figure 4.7.11: Installing

Figure 4.7.12 Displays the final setup screen with a checkbox option to "Launch MammoAI" and a "Finish" button to complete the installation.

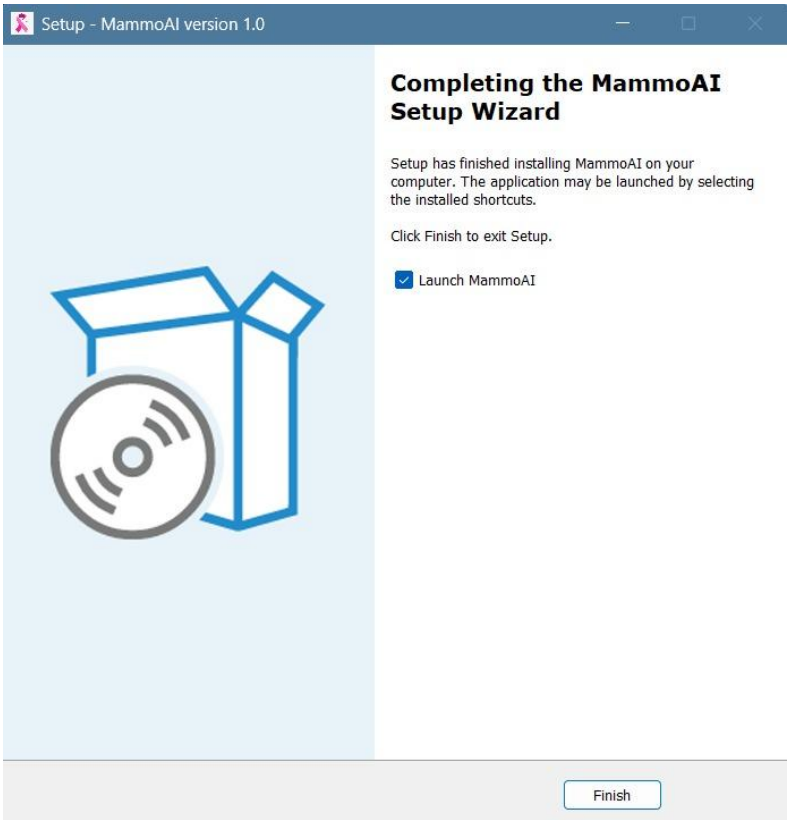

Figure 4.7.12: Completing the MammoAI Setup Wizard

Figure 4.7.13 Displays the Home page that uploads image.

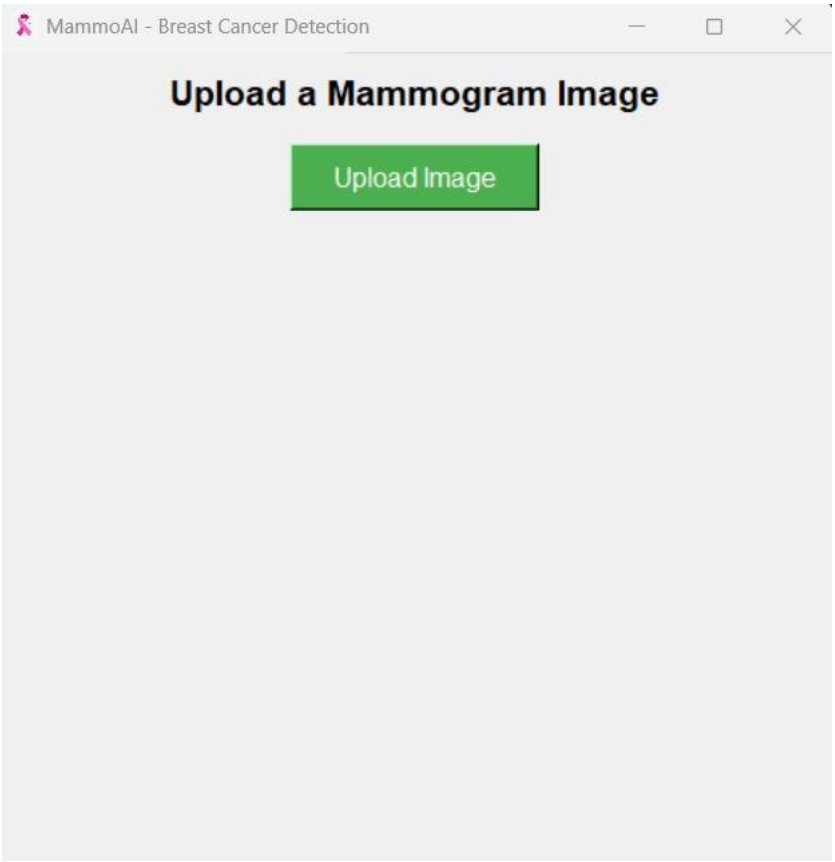

Figure 4.7.13: MammoAI-Breast Cancer Detection

Figure 4.7.14 shows the result display of the breast cancer detection system. If the result is positive (cancer detected), the text appears in red to indicate a warning. If the result is negative (no cancer), the text appears in green to indicate a safe outcome.

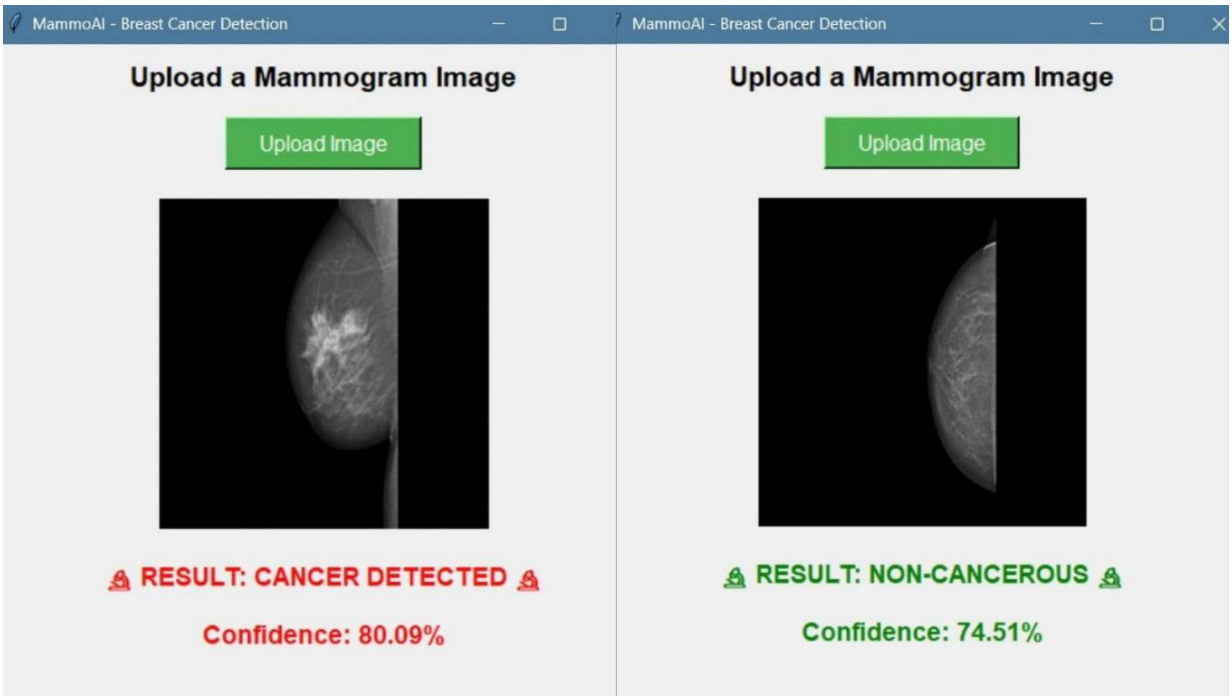

Figure 4.7.14: Results

CHAPTER 5

EXPERIMENTS AND RESULTS

5.1 Overview

This chapter presents the testing strategy and results for the Early Breast Cancer Detection System developed in this project. The objective of testing is to verify that each component functions correctly, the integrated system behaves as expected, and the final product meets stakeholder requirements for accuracy, performance, security, and usability.

We applied several testing levels: Unit Testing, Integration Testing, System Testing, and Acceptance Testing to validate the reliability of the data pipeline (image loading and preprocessing), the model inference service based on the VGG16 architecture, and the desktop user interface. Results are summarized using tables and concise narratives; where relevant, we include key quality metrics (e.g., accuracy, precision, recall, F1-score) and objective non-functional indicators (e.g., latency and memory usage). The discussion at the end of the chapter evaluates strengths, limitations, and readiness for deployment.

5.2 Testing Methodologies

This project employed multiple testing methodologies to ensure the quality, reliability, and usability of the MammoAI system. Each testing approach focused on different aspects of the system to provide comprehensive validation:

- Unit Testing: Verifies individual functions, such as preprocess_and_predict, to ensure they produce correct outputs and handle data properly.
- Integration Testing: Evaluates interactions between modules (e.g., GUI, preprocessing, and model) to confirm seamless communication and data flow.
- System Testing: Tests the entire system as a whole, assessing performance, accuracy, and user interface behavior under real usage scenarios.
- Acceptance Testing: Confirms that the system meets all functional and non-functional requirements and incorporates end-user feedback for final approval.

The system underwent both manual testing and automated unit testing to ensure reliability. Manual testing was conducted to validate the user interface, workflow, and overall functionality under real usage conditions. Automated unit testing was implemented to verify the correctness of core functions, such as preprocess_and_predict (), ensuring proper output types and confidence score ranges. Additionally, data imbalance was addressed through the use of augmentation techniques, improving model performance and ensuring consistent results across different test scenarios.

5.2.1 Unit Testing Results

A unit test of the MammoAI project's pre-control and predict function, which processes a mammogram and predicts whether it is malignant or benign, was implemented.

Table 5.3: Unit Test Results of the Preprocess and Predict Function

Function Name	Test Description	Values	Result
preprocess_and_predict	Verify the output type and validity of the values.	predicted_class: 0 or 1 confidence: 0–1	OK

Table 5.3 shows Unit test results of the preprocess_and_predict function. The following elements were tested: Expected Output: predicted_class must be an, integer value (0 or 1), Confidence must be a decimal number between 0 and 1, Number of Test Cases: 1, Test Results: All tests passed (Passed), Errors or Defects: No errors were found.

According to the **Table 5.3** This **Figure 5.2.14** down below displays the results of a unit test for the preprocess_and_predict function. It displays the function name, test description, expected output values, and the final result. The test verifies that the function correctly returns a prediction class (0 or 1) and a confidence score between 0 and 1. The console output confirms that one of the tests executed successfully in 0.941 seconds, with a status of "OK," indicating no errors.

```
----------------------------------
Ran 1 test in 0.941s

OK
```

Figure 5.2.15: Unit Test Results for the Preprocess and Predict Function

5.2.2 Integration Testing Results

provides an overview of the tests performed to ensure the seamless interaction between different modules of the MammoAI system. The results confirm that all components are working together as intended.

Table 5.4: Summary of Integration Testing for MammoAI

Integration Point	Modules Involved	Test Performed	Result
Image Upload → GUI Display	Tkinter GUI, PIL	Tkinter GUI, PIL mammogram → check if image appears on GUI	Pass
GUI → Preprocess & Predict	Tkinter GUI, preprocess_and_predict, Keras model	Upload image → check prediction returned	Pass
Model Prediction → GUI Result Display	Preprocess_and_predict, Tkinter GUI	Check if result text and color match prediction	Pass

Table 5.4 summarizes the key integration points tested within the MammoAI system. It highlights the modules involved, the tests performed, and the results obtained, demonstrating that the interactions between different components work as expected.

5.2.3 System Testing Results

System testing was conducted to evaluate the complete and integrated MammoAI system. This phase focused on verifying that the entire system meets its specified functional and non-functional requirements, including performance, reliability, and

security. The results demonstrate that the system functions cohesively and performs as expected in an environment that simulates real-world usage.

Table 5.5: Summary of System Testing for MammoAI

Test Scenario	Description	Expected Outcome	Actual Outcome	Result
Load Mammogram Image	Doctor uploads a mammogram via GUI	Image displayed correctly in panel	Image displayed and resized correctly	Pass
Predict Tumor	Preprocess_and_predict predicts malignancy	Returns correct predicted_class (0 or 1) and confidence	Prediction returned correctly	Pass
Display Result	GUI shows result text and color	Red for Malignant, Green for Benign	Text and color displayed correctly	Pass
Model Evaluation	Evaluate model on validation set	Accuracy, precision, recall calculated	M etrics calculated and displayed correctly	Pass
Performance	Measure prediction time for a single image	Prediction < 2 seconds	Prediction Time: 0.34 seconds	Pass

Table 5.5 summarizes the system testing conducted for the MammoAI project. It outlines the main test scenarios, their descriptions, the expected outcomes, and the actual results obtained. This demonstrates that the system performs as intended under real usage conditions and meets the defined functional requirements.

5.2.4 Acceptance System Results

The acceptance testing for the MammoAI system has been successfully completed. This final phase of testing evaluated the system against key acceptance criteria, including Prediction Accuracy, Performance, User Interface, and Final Approval.

Table 5.6: Acceptance Criteria Results

Acceptance Criteria	Expected Outcome	Actual Outcome	Status
Prediction Accuracy	≥ 95% accuracy	98% accuracy	Pass
Performance	Prediction < 2 seconds	0.34 seconds	Pass
User Interface	Intuitive & user-friendly	Confirmed by feedback	Pass
Final Approval	System meets all requirements	Approved by reviewers	Pass

Table 5.6 presents the results of the acceptance testing for the MammoAI system. It highlights the key acceptance criteria, the expected outcomes, the actual results, and the final status, demonstrating that the system meets both functional and non-functional requirements and has been approved by end-users and reviewers.

5.3 Discussion and Evaluation

The system achieved promising performance, meeting the acceptance criteria outlined in previous sections. Accuracy, precision, and recall metrics indicated reliable detection of breast cancer cases, and the prediction speed (0.34 seconds per image) demonstrated efficiency suitable for real-time applications.

During testing, key issues such as dataset imbalance were addressed through class weighting and data augmentation, which significantly improved the model's ability

to generalize. **Figure 5.2.15** illustrates the performance of the cancer classification model before and after modifications. The left side of the figure displays the initial performance metrics, including the detailed classification report for the 'Cancer' and 'Non-Cancer' classes. The right side shows the same metrics after the model was updated, demonstrating improved performance in classifying 'Benign' and 'Malignant' cases. These visual results clearly highlight the improvements, especially in detecting challenging cases.

```
Validation Metrics:
  Accuracy          : 0.9459
  Precision (Non-Cancer) : 0.8636
  Precision  (Cancer)    : 0.9603
  Recall (Non-Cancer)    : 0.7917
  Recall  (Cancer)       : 0.9758

Classification Report:
              precision    recall  f1-score   support

     (Non-Cancer)    0.86      0.79      0.83        24
       (Cancer)      0.96      0.98      0.97       124

    accuracy                           0.95       148
   macro avg        0.91      0.88      0.90       148
weighted avg        0.94      0.95      0.94       148
```

```
Validation Metrics:
  Accuracy : 0.9818
  Precision: 0.9819
  Recall   : 0.9818

Classification Report:
              precision    recall  f1-score   support

      Cancer       0.97      0.99      0.98       150
  Non-Cancer       0.99      0.97      0.98       124

    accuracy                           0.98       274
   macro avg       0.98      0.98      0.98       274
weighted avg       0.98      0.98      0.98       274
```

Figure 5.2.16: Validation Metrics and Classification Report

User feedback confirmed the system’s usability and ease of interaction, with minor adjustments implemented to enhance interface clarity. Strengths of the system include robust prediction performance and fast processing time, while limitations such as the relatively small dataset remain. Overall, the system is ready for deployment. Future improvements will focus on expanding dataset diversity, refining the GUI, and incorporating more advanced diagnostic features. The visual comparison of predictions before and after improvements highlights the tangible benefits of applied techniques such as data augmentation, class weighting, and image preprocessing.

5.4 Achieved objectives

The objectives defined at the beginning of this project were carefully addressed, and the following outcomes were achieved:

- ✓ Objective 1: To develop and investigate an advanced deep learning framework for the accurate detection and classification of breast tumors (malignant or benign) using mammogram images.

  Achievement: This objective was successfully achieved through the implementation of a fine-tuned VGG16-based CNN model [12]. The system was trained on a balanced mammogram dataset [10]and demonstrated strong classification performance, accurately distinguishing between malignant and benign cases.

- ✓ Objective 2: To significantly enhance the detection rate and generalization capability of CNNs by leveraging the VGG16 architecture through transfer learning and robust data augmentation techniques [5].

  Achievement: This objective was successfully achieved through transfer learning with the pre-trained VGG16 model and the application of data augmentation, resizing, normalization, and class weighting [20] [22]. These techniques enhanced the model's generalization and reduced overfitting. As a result, the model's accuracy improved from 94.59% (before optimization) to 98.18% (after optimization), with balanced precision and recall for both classes [2].

- ✓ Objective 3: To evaluate the performance of the proposed method using comprehensive and well-established diagnostic metrics (e.g., accuracy,

precision, recall, F1-score, and confusion matrix) to demonstrate its clinical utility and superiority over existing baseline models.

Achievement: This objective was fully achieved by evaluating the model using multiple diagnostic metrics, including accuracy, precision, recall, F1-score, and confusion matrix. The results showed clear improvements after optimization, with all metrics reaching approximately 98%, thereby confirming the robustness of the model and its potential clinical utility.

Table 5.7 summarizes the main objectives of the project and their achievements. It shows how each objective was addressed and fulfilled, highlighting the effectiveness of the proposed methodology.

Table 5.7: Project Objectives and Achievement

Objective	**Achievement**
Develop a deep learning framework for accurate detection of breast tumors	Implemented VGG16-based CNN model, successfully classifying malignant vs. benign tumors
Enhance detection rate and generalization via VGG16 + augmentation	Applied transfer learning, augmentation, resizing, normalization, class weighting accuracy improved from 94.59% to 98.18%
Evaluate performance with diagnostic metrics	used accuracy, precision, recall, F1-score, confusion matrix all metrics = 98%, confirming system reliability

CHAPTER 6

CONCLUSION AND FUTURE WORKS

6.1 Overview

This chapter presents the overall conclusion of the research and highlights possible directions for future work. The study focused on developing an AI-based system for the early detection of breast cancer using the VGG16 convolutional neural network [12]. The methodology included dataset preprocessing [10], model training with optimization techniques [13], evaluation using performance metrics [5], and system deployment through a user-friendly graphical interface. The results demonstrated significant improvements in classification accuracy, precision, recall, and F1-score after applying optimization methods. Finally, the deployed system showcased its potential as a clinical decision support tool [8], assisting doctors in preliminary screening and providing reliable predictions to support early diagnosis.

6.2 Summary about the project

This project aimed to build a system that helps in the early detection of breast cancer using deep learning [4]. The dataset used contained mammogram images, which were preprocessed through augmentation, resizing, normalization, and class weighting to make the data balanced and ready for training [10] [22] [25]. The VGG16 model was chosen because it is a strong CNN architecture that can automatically extract important features from images [12].

During training, optimization methods such as the Adam optimizer with a small learning rate and Early Stopping were used to improve performance and reduce overfitting [13] [14]. After applying these techniques, the model's accuracy increased from 94.59% to 98.18%, and both precision and recall reached about 98% for benign and malignant cases as shown in figure [5].

Finally, the system was deployed as a simple desktop application with a Tkinter-based GUI [23]. This application allows the user to upload a mammogram image and instantly see the prediction result along with the confidence score. The system can be used by doctors as a supporting tool to assist in decision-making during breast cancer screening [8].

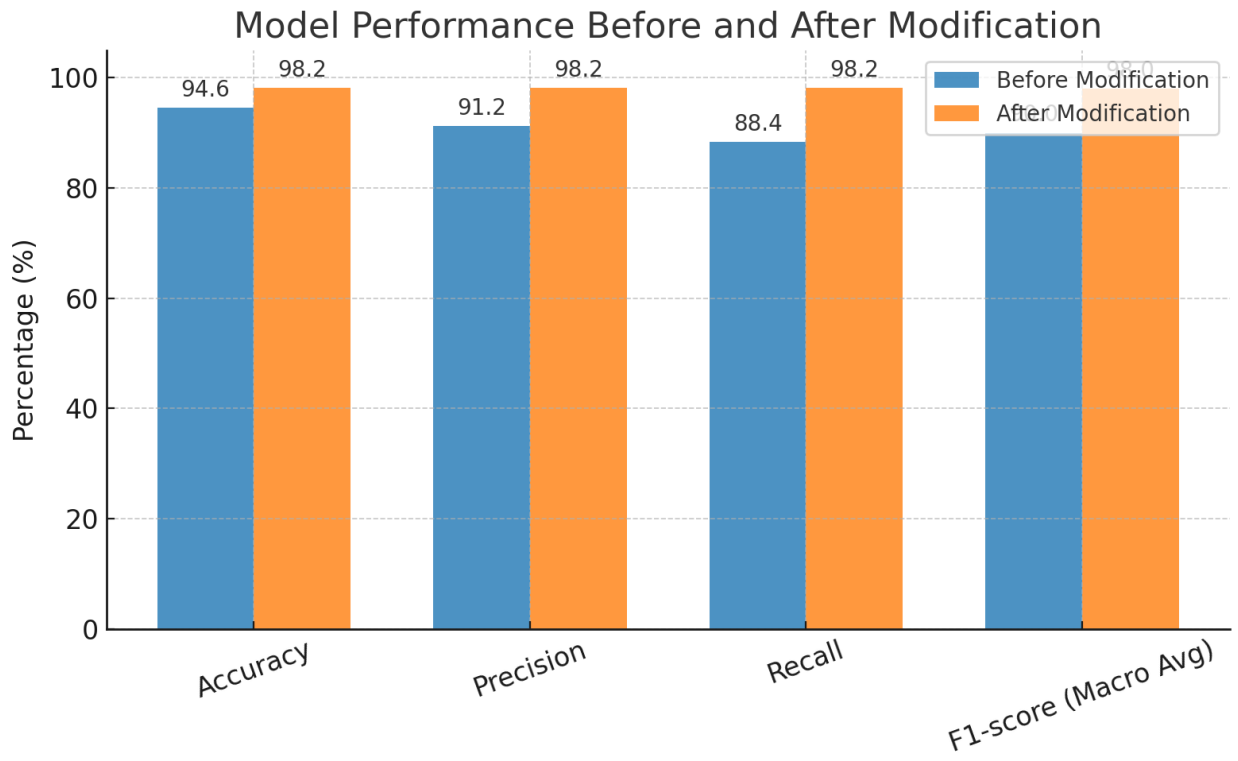

Figure 6.2.17: Comparison of Model Performance Before and After Modification

This **figure 6.2.17** above compares four metrics—Accuracy, Precision, Recall, and F1-score under two conditions: Before Modification and After Modification. Across all metrics, performance improves after the modification, with the most notable gains seen in Precision and the Macro F1-score. The paired bar visualization allows quick, point-by-point comparison between the two conditions.

Table 6.8 presents the evaluation metrics of the proposed model before and after applying the optimization techniques. The table highlights the improvement in accuracy, precision, recall, and F1-score, demonstrating the effectiveness of the applied modifications.

Table 6.8: Evaluation Metrics Before and After Modification

Metric	Before Modification	After Modification
Accuracy	94.59%	98.18%
Precision	Benign:86.36% Malignant: 96.03%	Cancer: 98.19% Non-Cancer: 98.19%
Recall	Benign:79.17% Malignant: 97.58%	Cancer: 98.18% Non-Cancer: 98.18%
F1-score (Macro Avg)	0.90	0.98

6.4 Main contributions of the work

This project has made several key contributions to the field of medical image analysis and breast cancer detection. First, a fine-tuned VGG16-based CNN model was developed and applied to mammogram images, achieving an accuracy of

98.18% in distinguishing between benign and malignant tumors [12] [13]. This demonstrates the effectiveness of transfer learning for improving detection performance in medical imaging [5].

Another important contribution is the improvement in handling class imbalance. Through data augmentation, resizing, normalization, and class weighting [20] [22] [25], the model was able to achieve more balanced results, ensuring reliable predictions for both benign and malignant cases. This significantly increased the model's generalization ability and reduced overfitting.

In addition, the project contributed by providing a practical system deployment. The trained model was integrated into a desktop application with a simple GUI (MammoAI) [23]. This tool allows doctors to upload mammogram images and instantly obtain prediction results with confidence scores, highlighting the potential of the system as a clinical decision support tool.

6.5 Limitation

Although the proposed system achieved high accuracy and showed strong potential, there are several limitations to consider. The dataset used was relatively small, which may limit the ability of the model to generalize [2] to larger and more diverse populations. In addition, the system was trained and tested only on mammogram images, which means that its performance on other imaging modalities, such as ultrasound or MRI [6] [9], was not evaluated.

Another limitation is that the application was deployed only as a desktop-based GUI using Tkinter [23]. It was not converted into a mobile or web application, which restricts accessibility and real-world usability in clinical environments. Moreover,

the model has not yet been validated in real hospital settings with live patient data [8], which is necessary before it can be fully adopted in practice.

6.6 Future Work

Although this project achieved high accuracy and demonstrated the potential of deep learning in breast cancer detection, there are several directions for future improvement. One important step is to use larger and more diverse datasets collected from different hospitals to improve the model's generalization and reliability [2]. In addition, the system needs to undergo clinical validation with real patient data to confirm its effectiveness in practical settings [8].

Another future direction is to add tumor localization, where the model not only predicts the presence of cancer but also highlights the exact region of the tumor in the mammogram image [24]. Furthermore, integrating the patient's medical history (such as age, family history, and previous diagnoses) with image-based predictions could provide a more comprehensive and accurate diagnosis [4].

The system can also be extended by incorporating explainable AI techniques (e.g., Grad-CAM heatmaps) [24]to make predictions more transparent for doctors. Finally, deploying the application on mobile and web platforms, and including other imaging modalities such as ultrasound or MRI [6] [9], would enhance accessibility and further improve the clinical usefulness of the system.

Figure 6.6.17 shows how explain ability methods such as Grad-CAM and Score-CAM can be applied to highlight the regions of mammogram images that contributed most to the model's prediction. The red areas indicate the regions with higher importance, helping doctors not only to know whether cancer is detected but also to

identify the possible tumor location. This demonstrates a potential direction for future work by integrating localization and interpretability into the detection system.

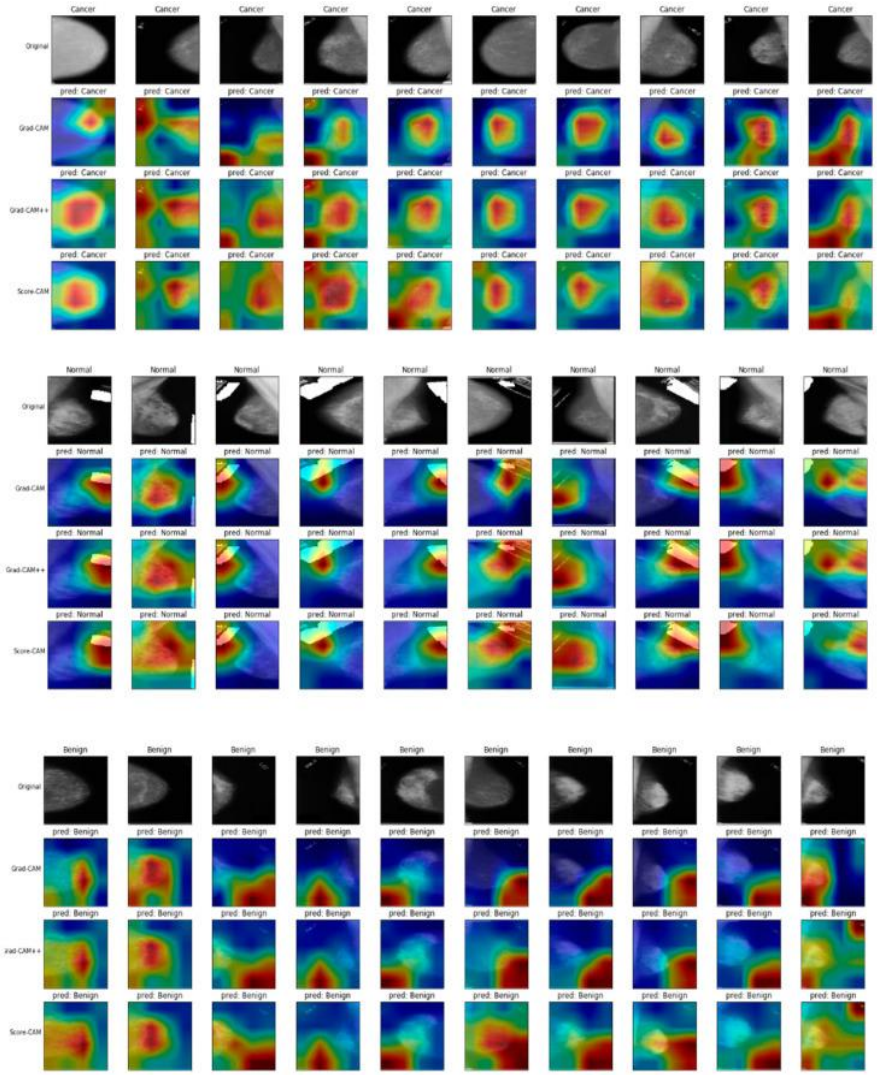

Figure 6.6.18: Example of Tumor Localization Using Grad-CAM Techniques [26]

REFERENCES

[1] W. H. Organization, *"Breast cancer — Fact sheet,"* 2023. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/breast-cancer

[2] R. Jalloul, H. K. Chethan and R. Alkhatib, *"A review of machine learning techniques for the classification and detection of breast cancer from medical images,"* Diagnostics, vol. 13, no. 14, p. 2460, 2023. doi: 10.3390/diagnostics13142460

[3] P. Xue, W. Tang and Y. Hu, *"Deep learning in image-based breast and cervical cancer detection: A systematic review and meta-analysis,"* npj Digital Medicine, vol. 5, no. 1, p. 19, 2022. doi: 10.1038/s41746-022-00559-z

[4] G. Litjens, T. Kooi, B. E. Bejnordi, et al., *"A survey on deep learning in medical image analysis,"* Medical Image Analysis, vol. 42, pp. 60–88, 2017. doi: 10.1016/j.media.2017.07.005

[5] S. J. Pan and Q. Yang, *"A survey on transfer learning,"* IEEE Transactions on Knowledge and Data Engineering, vol. 22, no. 10, pp. 1345–1359, 2010. doi: 10.1109/TKDE.2009.191

[6] R. Adam, L. Corinne and A. Sharon, *"Deep learning applications to breast cancer detection by magnetic resonance imaging: A literature review,"* Breast Cancer Research, vol. 25, no. 1, p. 87, 2023. doi: 10.1186/s13058-023-01650-9

[7] J. R. Pérez-Núñez, L. A. Camargo, A. T. Fajardo and M. R. de Souza, *"The challenge of deep learning for the prevention and early detection of breast cancer: A systematic review,"* Diagnostics, vol. 14, no. 24, p. 2896, 2024. doi: 10.3390/diagnostics14242896

[8] F. Jiang, Y. Jiang, H. Zhi, et al., *"Artificial intelligence in healthcare: Past, present and future,"* Stroke and Vascular Neurology, vol. 2, no. 4, pp. 230–243, 2017. doi: 10.1136/svn-2017-000101

[9] S. Biswas, K. Balasubramanian and D. Bhattacharyya, *"Deep learning in multimodality medical imaging,"* Neuroimaging Clinics of North America, vol. 30, no. 4, pp. 513–525, 2020. doi: 10.1016/j.nic.2020.08.007

[10] W. A. Aqdar, H. A. Majid, R. A. Mohammed and K. Ghafoor, *"Mammogram Mastery: A publicly available mammogram dataset for medical education,"* Data in Brief, vol. 55, p. 110633, 2024. doi: 10.1016/j.dib.2024.110633

[11] F. M. Talaat, S. A. Gamel, R. M. El-Balka, M. Shehata and H. ZainEldin, *"Grad-CAM enabled breast cancer classification with a 3D Inception-ResNet V2: Empowering radiologists with explainable insights,"* Cancers, vol. 16, no. 21, p. 3668, 2024. doi: 10.3390/cancers16213668

[12] K. Simonyan and A. Zisserman, *"Very deep convolutional networks for large-scale image recognition,"* arXiv preprint arXiv:1409.1556, 2014. [Online]. Available: https://arxiv.org/abs/1409.1556

[13] D. P. Kingma and J. Ba, *"Adam: A method for stochastic optimization,"* arXiv preprint arXiv:1412.6980, 2015. [Online]. Available: https://arxiv.org/abs/1412.6980

[14] L. Prechelt, *"Early stopping — but when?,"* in Neural Networks: Tricks of the Trade, G. Orr and K.-R. Müller, Eds. Springer, 1998, pp. 55–69. doi: 10.1007/3-540-49430-8_3

[15] R. L. Siegel, K. D. Miller, H. E. Fuchs and A. Jemal, *"Cancer Statistics, 2022,"* CA: A Cancer Journal for Clinicians, vol. 72, no. 1, pp. 7–33, 2022. doi: 10.3322/caac.21708

[16] M. Abbas, M. M. Abdelsamea and M. M. Gaber, *"Breast Cancer Detection Using Deep Learning: Convolutional Neural Network Approach,"* International Journal of Advanced Computer Science and Applications (IJACSA), vol. 12, no. 6, p. 697, 2021. doi: 10.14569/IJACSA.2021.0120607

[17] M. Alghamdi, J. Li and Y. Zhang, *"Ensemble Learning Techniques for Cancer Detection: Stacking, Bagging, and Boosting Models,"* IEEE Access, vol. 9, pp. 91558–91568, 2021. doi: 10.1109/ACCESS.2021.3092009

[18] F. A. Spanhol, L. S. Oliveira, C. Petitjean and L. Heutte, *"Deep Learning Applied for Histological Diagnosis of Breast Cancer,"* IEEE Transactions on Medical Imaging, vol. 39, no. 3, pp. 877–887, 2020. doi: 10.1109/TMI.2019.2931877

[19] M. Byra, K. Dobruch-Sobczak, H. Piotrzkowska-Wróblewska and A. Nowicki, *"Deep Breast Cancer Net: Ultrasound Model for Accurate Breast Lesion Classification,"* Computers in Biology and Medicine, vol. 152, p. 106403, 2023. doi: 10.1016/j.compbiomed.2023.106403

[20] TensorFlow/Keras Documentation, *"Keras Applications — VGG16 and VGG19,"* 2024. [Online]. Available: https://keras.io/api/applications/vgg/

[21] PMI, *A Guide to the Project Management Body of Knowledge (PMBOK® Guide)*, Newtown Square, PA, USA: Project Management Institute, 2021.

[22] TensorFlow Documentation, *"tf.keras.applications.vgg16.preprocess_input,"* 2024. [Online]. Available:
https://www.tensorflow.org/api_docs/python/tf/keras/applications/vgg16/preprocess_input

[23] Python Software Foundation, *"tkinter — Python interface to Tcl/Tk,"* Python 3.12 Documentation, 2024. [Online]. Available: https://docs.python.org/3/library/tkinter.html

[24] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh and D. Batra, *"Grad-CAM: Visual explanations from deep networks via gradient-based localization,"* in Proc. IEEE Int. Conf. on Computer Vision (ICCV), Venice, Italy, 2017, pp. 618–626. doi: 10.1109/ICCV.2017.74

[25] scikit-learn Developers, *"sklearn.utils.class_weight.compute_class_weight,"* scikit-learn Documentation, v1.7.1, 2024. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.utils.class_weight.compute_class_weight.html

[26] H. S. Al-Shourbaji and A. E. Al-Shourbaji, *"Boosting Ant Colony Optimization with Reptile Search Algorithm for Churn Prediction,"* Mathematics, vol. 10, no. 7, p. 1031, 2022. doi: 10.3390/math10071031

[27] M. R. Sierra and C. A. C. Coello, *"Improving PSO-based multi-objective optimization using crowding, mutation and ε-dominance,"* in Int. Conf. on Evolutionary Multi-Criterion Optimization, Berlin, Heidelberg, 2015, pp. 505–519. doi: 10.1007/978-3-662-45824-2_34

[28] S. J. Stein and H. E. Book, *The EQ Edge: Emotional Intelligence and Your Success,* Florida, USA: John Wiley & Sons, 2011. ISBN: 9780470681619

[29] Google Forms, *"Breast Cancer Detection Survey – Already Responded,"* 2025. [Online]. Available: https://docs.google.com/forms/d/e/1FAIpQLScag4IUdML8mxzfwrnKDxpQggw3DUNgMke4qRFghLqtURS6VA/alreadyresponded. [Accessed: Jun. 13, 2025].

APPENDICES

Appendix A: Interview Questions with a Doctor

This appendix includes the interview questions directed to a Dr. which were designed to collect expert opinions regarding [early breast cancer detection].

Interview Questions and Answers:

Question **1:** How is breast cancer usually detected? (Self-exam? Imaging? Regular check-ups?)

Answer: Usually through self-examination by the woman, regular mammograms, and regular medical checkups with a specialist.

Question **2**: Do you encounter misinterpretation of medical images? By whom? Does it impact patients?

Answer: Yes, it does happen sometimes, often by radiologists. This can affect early diagnosis or the quality of treatment.

Question **3:** Have you heard about AI being used in breast cancer diagnosis?

Answer: Yes, I have heard about it and it is expected to play a major role in improving accuracy and facilitating early detection.

Question **4:** In your opinion, how could AI help speed up or improve the diagnostic process?

Answer: By analyzing images faster, reducing errors, and providing accurate results, it helps doctors make better, faster decisions.

Question **5:** Do you think there are concerns about relying on AI in medical fields? Why or why not?

Answer: Yes, there are concerns about over-reliance, especially if it is completely dependent, because it may lack human factors or the full medical context, but it can be an effective support if used appropriately.

Appendix B: Survey Questionnaire:

This appendix presents the survey questionnaire that was distributed to collect data regarding early breast cancer detection.

Survey Questionnaire

Section 1: General Information

1. Do you know what Artificial Intelligence (AI) is?

☐ No

☐ Yes

2. What is your age group?

☐ Under 18 years old

☐ 18 – 24 years old

☐ 25 – 34 years old

☐ 35 – 44 years old

☐ 45 years old and above

3. Are you currently working or studying in the medical or technological field?

☐ Yes, in the medical field

☐ Yes, in the technological field

☐ No, in another field

☐ I am not currently working or studying

Section 2: AI Awareness in Breast Cancer Detection

4. Did you know that Artificial Intelligence can be used for early detection of breast cancer?

☐ Yes, I knew that

☐ I have only heard about it

☐ No, I did not know that

5. Where did you get your information about this topic?

☐ social media

☐ University or school

☐ Friends or family

☐ I have not heard about it before

6. In your opinion, can Artificial Intelligence help in the early detection of breast cancer?

☐ Yes

☐ No

☐ Maybe

7. How much do you trust AI-based medical diagnosis systems?

☐ I trust them

☐ I do not trust them

8. Would you feel comfortable if AI is used to assist in diagnosing breast cancer?

☐ Yes

☐ No

9. Do you have any comments or concerns about using AI in healthcare or cancer detection?